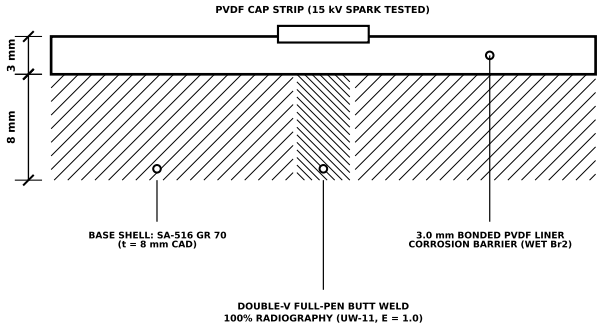
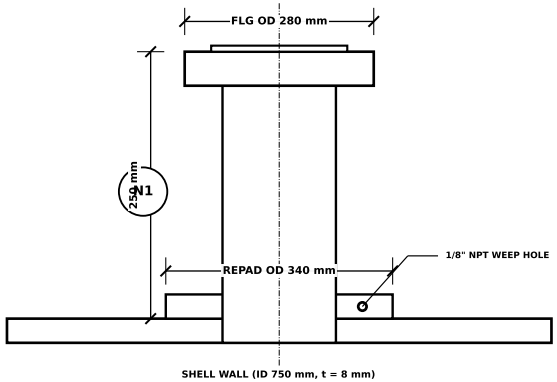
DETAIL 1 — ASME SEC VIII SHELL WELD & LINER

TYPE 1 BUTT WELD & 3 mm PVDF LINER — SCALE 1:4



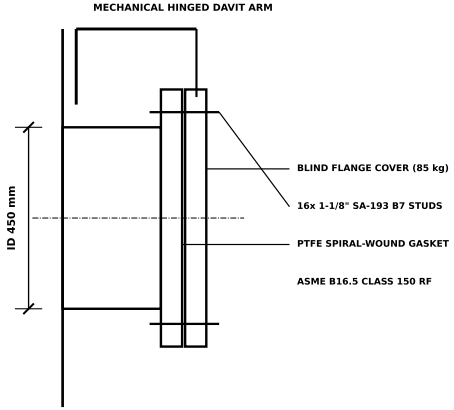
DETAIL 2 — NOZZLE N1 (DN150) & REPAD

ASME B16.5 WELD NECK & UG-37 REPAD — 1:10



DETAIL 3 — ACCESS MANWAY (DN450) & DAVIT

ASME B16.5 BLIND COVER & HINGED DAVIT — 1:12



DETAIL 4 — LIFTING LUG & EARTHING BOSS

RIGGING PADEYE (5.0 t SWL — PRELIM) & GROUNDING — 1:10

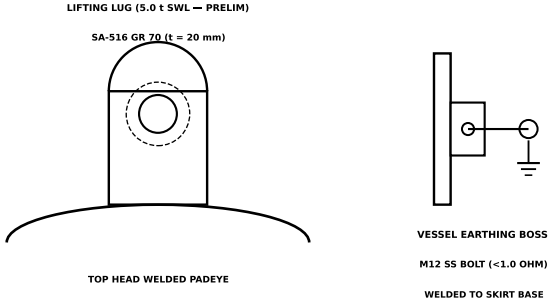


TABLE 1 — NOZZLE SCHEDULE & REPADS

NOZ	SERVICE	SIZE	FLG OD	REPAD
N1	Feed Gas Inlet	DN 150	280 mm	DIA 340 mm
N3	Water Scrub Inlet	DN 80	190 mm	DIA 190 mm
N4	Rich Liquor Outlet	DN 80	190 mm	Integral
N5	Shell Bottom Drain	DN 40	127 mm	None
N6	Treated Gas Vent	DN 50	152 mm	None
M1	Lower Access Manway	DN 450	635 mm	**DIA 750 mm**
M2	Upper Access Manway	DN 450	635 mm	**DIA 750 mm**

TABLE 2 — WELDING & NDE SCHEDULE

JOINT	LOCATION	WELD TYPE	NDE METHOD
Cat. A	Longitudinal Shell	Double-V Butt	100% Radiography (RT)
Cat. B	Girth Seams	Double-V Butt	100% Radiography (RT)
Cat. C	Flange Hubs	Full Pen Hub	100% Dye Penetrant
Cat. D	Nozzle Necks	Set-Through	100% PT + Ultrasonic
Repad	Reinforcing Pads	Continuous	Air Leak Test (1.5 bar)
Liner	PVDF Membrane	Thermal Cap	15 kV Spark Testing

PRESSURE VESSEL FABRICATION NOTES

1. CODE JURISDICTION: PRELIMINARY DESIGN PER ASME SEC VIII DIV 1.
2. LINING: 3 mm SOLID PVDF TESTED AT 15 kV (NACE SP0188).
3. HYDROSTATIC TEST PRESSURE: AS PER ASME VIII DIV. 1 (CHLORIDE-FREE DM WATER).

JADAVPUR UNIVERSITY | CHEMICAL ENGINEERING DEPARTMENT | PROCESS EQUIPMENT DESIGN & DRAWING

DRAWING TITLE	LAB / SUBJECT	COURSE / BRANCH	SCALE
PACKED-BED ABSORBER (C-101) — MECHANICAL FABRICATION DETAILS CAD	PROCESS EQUIPMENT DESIGN & DRAWING	B.E. CHEMICAL ENGINEERING	DETAILED 1:4, 1:10 METRIC
STUDENT NAME	**ROLL NO.**	**DRAWING NO.**	**DATE**
SOURISH SENAPATI	002310301035	PEDD-PBA-04	29 / 09 / 2026

PACKED ABSORBER (C-101) | 0.25 m3/s GAS (2% Br2) | WATER: 17.25 kg/s | ASME SEC VIII DIV 1 / B16.5